

Ghaf crown mapping

Quick reference

Everything here is expanded in the technical manual; the numbers in the right-hand column point at its chapters. Values come from `docs/handover/FACTS.yml`, which wins over this sheet if they ever disagree. Every command below assumes the `ghaf` environment is active, the working directory is the code folder, and `MODEL` names the FastViT model folder.

Prove it works — in this order

Command	Passes when	Ch.
<code>python -m pytest tests\ -q</code>	339 passed and no F — the code is intact. The number skipped varies with the machine	4
<code>python tools\smoke_test.py --checkpoints ../models</code>	six rows of ok, every tensor matched	4
<code>python tools\check_dataset.py ../data\ghaf --sample 0</code>	7005 / 869 / 767 paired	4

0.00% `ghaf` in the prediction column of the second command is correct: that forward pass runs on a blank synthetic tile, not on imagery.

Map one orthomosaic

```
python -m ghaf.inference.large_image ^
%MODEL%\fastvit-ma36_mask2former.py ^
%MODEL%\best_mIoU_iter_3500.pth ^
..\samples\Kalba26_sample.tif ^
--out-mask ../output\crowns.tif ^
--out-polygons ../output\crowns.gpkg ^
--batch-size 4 --min-area 1
```

THE DEFAULTS ARE NOT THE SETTINGS THAT PRODUCED THE RESULTS

Batch size defaults to 1, and `--min-area` to 0.0 — which keeps every single-pixel fragment as a crown and inflated the count on the sample clip from 16 to 27. `--out-polygons` will not run without `--out-mask` or `--out-prob`, and `--tile` **must stay 1024**: change it and the run still succeeds, still writes all three outputs, and reports a canopy figure produced at the wrong scale. (*Manual §6.5*)

Map a folder of images

```
python tools\predict_folder.py ^
%MODEL%\fastvit-ma36_mask2former.py ^
%MODEL%\best_mIoU_iter_3500.pth ^
D:\my-images --out-dir ../output\folder ^
--batch-size 4 --min-area 1 --limit 3
```

Drop `--limit 3` once three images have come out right. One unreadable file does not stop the run: it is recorded in `summary.json` and the exit status is non-zero, neither of which is visible to somebody watching the progress bar. (*Manual §7.2*)

The numbers

	Value		Value
Tile	1024 px	Trained GSD	2.68 cm/px
Overlap	512 px	Tile on the ground	27.44 m (0.075 ha)
Threshold	0.5	Scratch needed	9 bytes per source pixel
Batch used here	4	Sample clip	8192 × 8192, 225 windows
Canopy, clip	0.9687 %	Canopy, test split	3.44 %
Crowns, clip	27 raw, 16 filtered	Crown area	2.4 to 112 m²

Deployed model mIoU **79.32**, F1 **87.22** on `testing/ghaf26`. Identify a checkpoint by its digest, not its file-name: `f26cd5257b55058f...` for `best_mIoU_iter_3500.pth`, 252,650,755 bytes. The full digests are in `ghaf/release.py`.

Error → cause

What you see	Actually means	Ch.
No module named <code>mmengine</code>	wrong Python; <code>conda activate ghaf</code>	13
<code>mmseg ...</code> is not installed in conda environment	wrong environment; the message names the interpreter	13
No module named <code>ftfy</code>	<code>pip install ftfy regex</code>	13
Numpy is not available	NumPy 2 beside torch 1.12; <code>pin <2</code>	13
CUDA out of memory	lower <code>--batch-size</code> ; never <code>--overlap</code>	13
not enough scratch space	<code>--scratch-dir</code> at a bigger disk	13
<code>--out-polygons</code> needs <code>--out-mask</code> or <code>--out-prob</code>	polygons are traced from a written raster	13
<code>path</code> specified printed twice	an <code>&</code> in the path; quote it	13
<code>no such folder ... D:\ghaf-project</code>	the manual's example path; use your own	13
SHA-256 mismatch	that checkpoint is not the released file	13
<code>0.00% ghaf</code> from <code>smoke_test.py</code>	correct; a blank tile	4
<code>0.00%</code> canopy on real imagery	wrong checkpoint, or bands not RGB	13
a crown count far too high	fragments; use <code>--min-area 1</code>	13
high mIoU, predicts nothing	masks encoded 0 and 255, not 0 and 1	13

Rules that are not negotiable

Rule	Because
Never edit <code>configs/_base_/ghaf.py</code>	all six models inherit it; a change silently redefines what every published score means
Keep <code>work_dirs</code> logs	no seed is set, so the seed of a run exists only in its log
<code>--load-from</code> to fine-tune, never <code>--resume</code>	<code>--resume</code> continues a schedule; <code>--load-from</code> starts a new one
Score a fine-tuned model on the old split too	it can gain on the new site and lose on the original, and nothing reports it
Keep companion files beside their tiles	moving a PNG without its <code>.pgw</code> strips the position silently
<code>check_dataset.py</code> before any training run	masks encoded 0 and 255 train to a high mIoU and predict nothing

Never established, so do not quote it: how long a full-mosaic run takes — the 84,072 × 103,691 mosaic has never been run end to end, and its 78.5 GB of scratch and 33,128 windows are arithmetic. Also unmeasured: training time, validation scores for the five models other than FastViT-MA36, and how often touching crowns merge into one polygon. (*Manual §14.3*)